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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/096,681	11/12/2020	Michael W. Olszowy	1160015USP	8384

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EXAMINER
KINSEY WHITE, NICOLE ERIN

ART UNIT	PAPER NUMBER
1648	

NOTIFICATION DATE	DELIVERY MODE
01/24/2024	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/096,681	Applicant(s) Olszowy, Michael W.	
	Examiner Nicole Kinsey White	Art Unit 1648	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 01 September 2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-2,4-11,13-15 and 19-27 is/are pending in the application.

5a) Of the above claim(s) 5 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-2,4,6-11,13-15 and 19-27 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Withdrawal of Finality of Last Office Action

Applicant's request to withdraw the finality of the last Office Action dated December 6, 2023 is persuasive and, therefore, the finality of that action is withdrawn.

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(b):
(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claim 15 is rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

Claim 15 is confusing. It is not clear what is meant by the phrase “calculated from the analytically quantified number concentration of the viral standard solution used to make the evaluation solution”. Accordingly, one of ordinary skill in the art will not know the metes and bounds of the claim. The claim should be rewritten to clarify what is being claimed.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to

point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claims 1-2, 7-11, 13-14 and 19-27 are rejected under 35 U.S.C. 103 as being unpatentable over Leisi et al. (Journal of Membrane Science, March 5, 2020, 603:118012, pages 1-14) and further in view of Kayukawa et al. (U.S. Patent Application No. 2023/0103302; effectively filed February 28, 2020) and Rockey et al. (Current Opinion in Biotechnology 2019, 57:42–49).

The instant claims are directed to a method for viral clearance evaluation, the method comprising:

(a) subjecting an evaluation solution to purification processing to prepare a treated solution, wherein the evaluation solution comprises fluorescent viral particles, and wherein the purification processing comprises at least one viral removal technique for separation of a viral contaminant; and

(b) determining a degree of removal of the fluorescent viral particles between the evaluation solution and the treated solution;

wherein the fluorescent viral particles comprise viral particles and a fluorescent dye conjugated to the viral particles;

wherein the determining a degree of removal of the fluorescent viral particles comprises subjecting a sample of the treated solution to an analytical technique to analytically quantify a residual concentration of the fluorescent viral particles in the

treated solution from fluorescent properties of the fluorescent viral particles, and the analytical technique comprises flow cytometry.

Leisi et al. teaches that virus filtration is a highly effective method in the downstream processing of biotherapeutic products. Leisi et al. discloses methods and standard solutions for viral clearance evaluation of parvovirus from PBS, comprising employing known concentrations of different fluorescently labeled parvovirus particles (MVM and CPV) spiked into PBS in filtration experiments [claim 1, part (a), claim 14, claim 26, claim 27] and assessing the filtration efficacy by detecting the labeled viruses inside the filter membrane using laser scanning microscopy.

Leisi et al. does not teach the use of flow cytometry detect residual fluorescent viral particles. However, Kayukawa et al. teaches a method for evaluating viral clearance of a biological material. Kayukawa et al. states "to more accurately evaluate the viral clearance capability of a virus removal medium, a viral quantification method capable of measuring a total viral capsid present in the biological material-containing solution can be selected, or a quantification method can be appropriately selected depending on the type of the virus used in the viral clearance capability evaluation. Examples of the viral quantification method include quantitative PCR, fluorescence flow cytometry, a $TCID_{50}$ method, and a plaque method (see paragraph [0131]). Kayukawa et al. further teaches that nucleic acid from the virus can be quantified by quantitative PCR or fluorescence flow cytometry (see paragraphs [0045], [0135] and [0140]). Rockey et al. teaches spiking water with fluorescently labeled viral particles to monitor and accurately validate viral removal. Residual viral particles were detected using flow virometry.

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the methods taught by Leisi et al. and use methods known in the art to detect the presence of the fluorescent viral particles in viral clearance evaluations. One of ordinary skill in the art may use laser scanning microscopy as taught by Leisi et al., virus-specific qPCR as taught by Leisi et al., quantitative PCR as taught by Kayukawa et al., fluorescence flow cytometry as taught by Kayukawa et al. or fluorescent flow virometry as taught by Rockey et al. and the result would be predictable (detection (or no detection) of viral particles in the filtrate).

For claim 2, Leisi et al. teaches that a filter membrane is used to separate the labeled virus from the solution.

For claims 7 and 8, Leisi et al. teaches that MVM or CPV parvoviruses can be used in the disclosed virus removal assay. Thus, one of ordinary skill in the art would understand that either virus can be used separately in different solutions or the viruses can be used together in the same solution with each virus labeled with a different fluorescent dye. Further, it would be obvious to one of ordinary skill in the art to use the method disclosed by Leisi et al. on a second, third, etc. solution comprising a biotherapeutic different from the first solution.

For claim 9, Leisi et al. teaches that virus filtration is highly effective in the downstream processing of biotherapeutics. It is known that biotherapeutics are derived from various means such as cell culture, chemical synthesis, etc. Thus, contaminants, such as cellular debris or unreacted chemicals would be present.

For claims 10, 13, 26 and 27, Leisi et al. teaches that virus filtration studies are commonly based on measurements of the viral load reduction between feed and filtrate

and expressed as the logarithmic reduction value (LRV) of the virus titer. The LRV provides a reliable information regarding the removal of the infectious virus and hence is used to validate the pathogen safety of biotherapeutics. Regarding the technique to quantify the fluorescent viral particles, it is well within the purview of one of ordinary skill in the art to use methods well-known in the art such as flow cytometry, scanning electron microscopy, fluorescence microscopy or transmission electron microscopy (see the Introduction) to quantify fluorescent viral particles in the feed and/or in the filtrate and/or in the retentate and the result would be predictable (detection/quantification of the fluorescent viral particles). In addition, it is obvious for and well within the purview of one of ordinary skill in the art to use the same technique for each sample taken (e.g., feed, filtrate and retentate) for consistency.

For claims 11 and 23-25, Leisi et al. does not use the disclosed method on a solution comprising a medical product. However, in view of the teachings of Leisi et al.[1], it would be obvious for one of ordinary skill in the art to use the method disclosed by Leisi et al. to purify solutions with biological medical products.

For claims 19, Leisi et al. does not teach that the parvoviruses are non-infectious or non-replicating. However, it would be obvious for one of ordinary skill in the art to use labeled non-infectious particles (e.g., LVPs) in the assay of Leisi et al. for safety reason and the result would be predictable (detection of labeled VLPs in the filter membrane).

For claim 20, Leisi et al. used parvoviruses.

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[1] Leisi et al. teaches that mammalian cells used for the production of recombinant proteins such as monoclonal antibodies are known to spontaneously express endogenous retrovirus-like particles. Adventitious contaminations of viruses on the other hand can be introduced into the cell culture by raw materials (see Introduction).

For claim 21, Leisi et al. demonstrates that the modification of the viruses with a **limited number of fluorophores did not significantly change the size or surface electrostatics of the capsids** (see Supplementary Figure 1). Supplementary Figure 1A shows the diameter of labeled MVM is approximately 1-3 nm larger than the diameter of unlabeled MVM. Leisi et al. found that the comparison of non-labeled with labeled viruses revealed no significant increase of the hydrodynamic diameter of the capsids, which reflects the apparent size of the hydrated viral particles in solution, even with the largest dyes attached (see paragraph bridging pages 7-8).

Claims 4 and 6 are rejected under 35 U.S.C. 103 as being unpatentable over Leisi et al. (Journal of Membrane Science, March 5, 2020, 603:118012, pages 1-14) as applied to claims 1-2, 7-11, 13-14 and 19-21 above, and further in view of Strauss et al. (Biotechnology and Bioengineering, 2009, 104(2):371-380).

The instant claims are directed to the method of claim 1 where the evaluation solution is treated with different viral removal techniques and each viral removal technique is assayed for the degree of viral removal.

The teachings of Leisi et al. are outline above and incorporated herein. Leisi et al. does not teach combining different virus removal techniques. Strauss et al. teaches that during production of therapeutic monoclonal antibodies (mAbs) in mammalian cell culture, it is important to ensure that viral impurities and potential viral contaminants will be removed during downstream purification. Anion exchange chromatography provides a high degree of virus removal from mAb feedstocks (see the Abstract).

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the methods taught by Leisi et al. and

include additional virus removal techniques. One would have been motivated to do so to increase purity of the biotherapeutic and to eliminate virus contaminants. There would have been a reasonable expectation of success in doing so given the teachings of Leisi et al. [the manufacture of biotherapeutic products includes dedicated process steps to inactivate or remove potential viral contaminants] and Strauss et al. [during production of therapeutic monoclonal antibodies (mAbs) in mammalian cell culture, it is important to ensure that viral impurities and potential viral contaminants will be removed during downstream purification].

Further, the courts have said: "It is prima facie obvious to combine two compositions each of which is taught by the prior art to be useful for the same purpose, in order to form a third composition to be used for the very same purpose . . . . [T]he idea of combining them flows logically from their having been individually taught in the prior art." In re Kerkhoven, 626 F.2d 846, 850, 205 USPQ 1069, 1072 (CCPA 1980) (citations omitted) (Claims to a process of preparing a spray-dried detergent by mixing together two conventional spray-dried detergents were held to be prima facie obvious.) See also In re Crockett, 279 F.2d 274, 126 USPQ 186 (CCPA 1960) (Claims directed to a method and material for treating cast iron using a mixture comprising calcium carbide and magnesium oxide were held unpatentable over prior art disclosures that the aforementioned components individually promote the formation of a nodular structure in cast iron.); and Ex parte Quadranti, 25 USPQ2d 1071 (Bd. Pat. App. & Inter. 1992) (mixture of two known herbicides held prima facie obvious). (See MPEP §2144.06(I) – Combining Equivalents Known For The Same Purpose).

It is the Examiner's position that MPEP §2144.06(I) also applies to combining methods each of which is taught by the prior art to be useful for the same purpose. In this case, the instant claims are combining known virus removal techniques for the purpose of removing viruses from a biotherapeutic solution.

Therefore, the invention as a whole is prima facie obvious to one of ordinary skill in the art at the time the invention was made.

Claim 22 is rejected under 35 U.S.C. 103 as being unpatentable over Leisi et al. (Journal of Membrane Science, March 5, 2020, 603:118012, pages 1-14) as applied to claims 1-2, 7-11, 13-14 and 19-21 above, and further in view of AAT Bioquest (Safety Data Sheet, Revision Date March 27, 2023).

The instant claim is directed to the method of claim 1 where the fluorescent dye is conjugated to viral protein of the viral particles and the fluorescent dye is in a moiety including a residual nitrogen of a reacted primary amine of the viral protein, and wherein the moiety has a molecular mass of no larger than 2,500 Daltons.

Regarding claim 22, the fluorescent dye, which is an amine-reactive N-hydroxy-succinimide ester-modified Atto dye, was attached to the capsid protein of the MVM (see section 2.7). As evidenced by AAT Bioquest, the Atto-488 dye used to label MVM has a molecular mass of about 788.

Response to Arguments

In the reply dated September 1, 2023, applicant first argues that the cited references do not teach using flow cytometry to quantify residual viral particles. Applicant's arguments have been fully considered and not found persuasive.

In view of the teachings of Leisi et al., Kayukawa et al. and Rockey et al., it would have been obvious for one of ordinary skill in the art to modify the methods taught by Leisi et al. and use methods known in the art to detect the presence of the fluorescent viral particles in viral clearance evaluations. One of ordinary skill in the art may use laser scanning microscopy as taught by Leisi et al., virus-specific qPCR as taught by Leisi et al., quantitative PCR as taught by Kayukawa et al., fluorescence flow cytometry as taught by Kayukawa et al. or fluorescent flow virometry as taught by Rockey et al. and the result would be predictable (detection (or no detection) of residual viral particles in the filtrate).

As for the analysis of fluorescent particles in the feed solution or filtrate solution, Leisi et al. teaches that virus filtration studies are commonly based on measurements of the viral load reduction between feed and filtrate and expressed as the logarithmic reduction value (LRV) of the virus titer. The LRV provides a reliable information regarding the removal of the infectious virus and hence is used to validate the pathogen safety of biotherapeutics. One of ordinary skill in the art would readily understand that viral load can be measured via detecting the fluorescent label of the virus, detecting viral proteins, or detecting viral nucleic acid.

Conclusion

No claim is allowed.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Nicole Kinsey White whose telephone number is (571)272-9943. The examiner can normally be reached M to Th 6:30 am to 4:30 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Janet Andres can be reached on 571-272-0867. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/NICOLE KINSEY WHITE/
Primary Examiner, Art Unit 1648